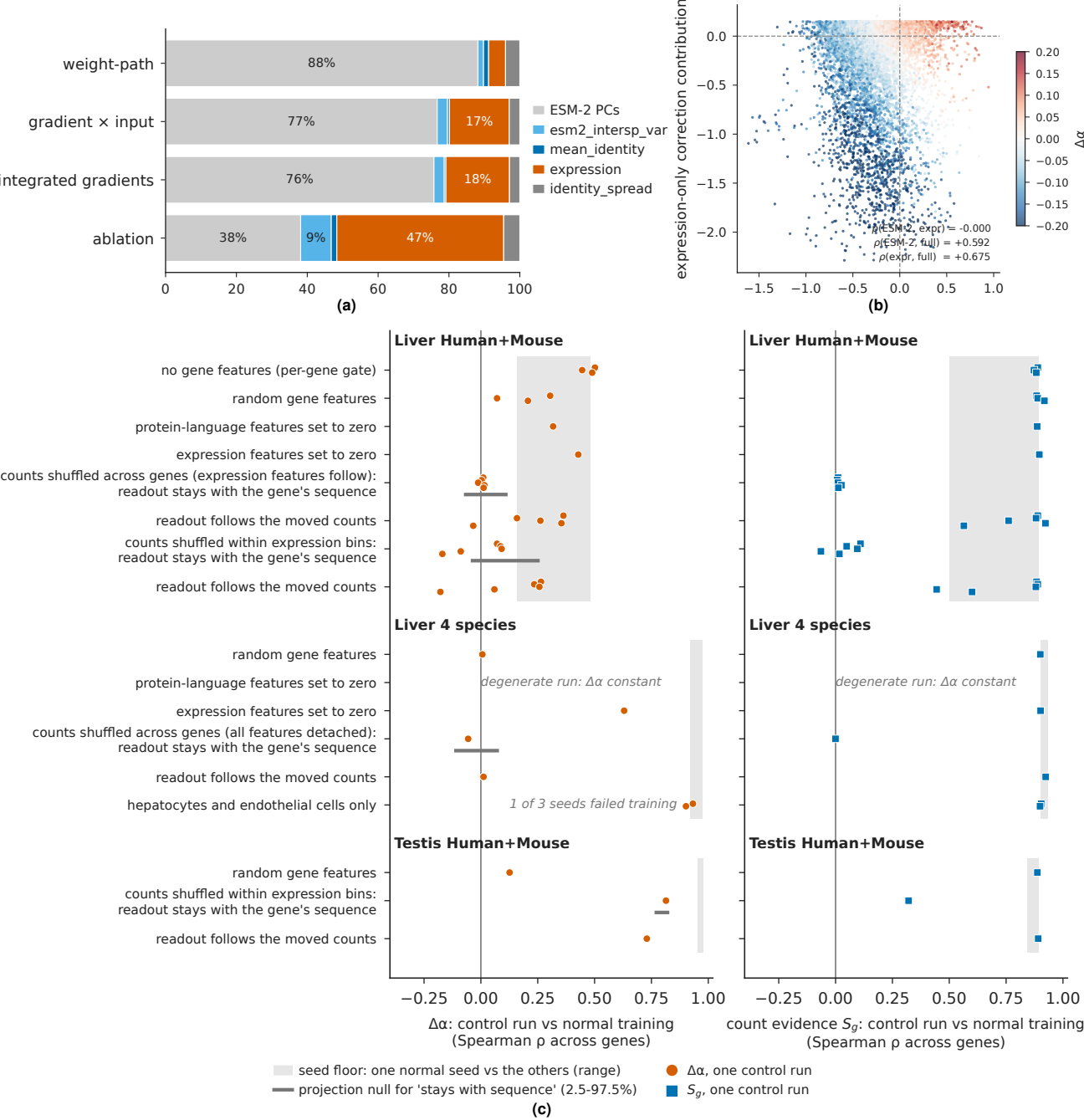

weight-path
gradient × input
integrated gradients
ablation
88%
77%
17%
76%
18%
38%
9%
47%
0
20
40
60
80
100
(a)
ESM-2 PCs
esm2_intersp_var
mean_identity
expression
identity_spread
expression-only correction contribution
Δα
ρ(ESM-2, expr) = -0.000
ρ(ESM-2, full) = +0.592
ρ(expr, full) = +0.675
(b)
Liver Human+Mouse
no gene features (per-gene gate)
random gene features
protein-language features set to zero
expression features set to zero
counts shuffled across genes (expression features follow): readout stays with the gene's sequence
readout follows the moved counts
counts shuffled within expression bins: readout stays with the gene's sequence
readout follows the moved counts
Liver 4 species
random gene features
protein-language features set to zero
degenerate run: Δα constant
expression features set to zero
counts shuffled across genes (all features detached): readout stays with the gene's sequence
readout follows the moved counts
hepatocytes and endothelial cells only
1 of 3 seeds failed training
Testis Human+Mouse
random gene features
counts shuffled within expression bins: readout stays with the gene's sequence
readout follows the moved counts
Δα: control run vs normal training (Spearman ρ across genes)
count evidence S_g: control run vs normal training (Spearman ρ across genes)
seed floor: one normal seed vs the others (range)
projection null for 'stays with sequence' (2.5-97.5%)
Δα, one control run
S_g, one control run
(c)